

# Quality Inspection Report: Selective Laser Melting of 316L Stainless Steel Flat Washers

**Report Number:** QIR-AM-316L-231026

**Date of Inspection:** October 26, 2023

**Inspector:** Maria Rodriguez, Senior Quality Engineer

**Batch Identification:** SLM-BATCH-WSH-231025-33

**Customer Reference:** Internal Quality Assurance Program

**Applicable Standards:** ASTM F3184-16, ISO/ASTM 52900, Internal Quality Procedure QP-AM-07

## Executive Summary

This report documents the comprehensive quality inspection for the manufacturing lot of 316L stainless steel flat washers produced via Selective Laser Melting (SLM) technology. The inspection was conducted to verify compliance with material specifications, dimensional requirements, and process control parameters. All 33 finished washers met the acceptance criteria with no non-conformances identified. The manufacturing process demonstrated stable operation with material usage and energy consumption within expected parameters for this production cycle.

**Overall Inspection Status: PASS**

## 1.0 Introduction and Scope

This quality inspection covers the complete manufacturing cycle of 316L stainless steel flat washers produced using Selective Laser Melting technology. The inspection scope includes:

- Verification of input material specifications and handling procedures
- Monitoring and recording of process parameters during both powder production and SLM operations
- Dimensional and visual inspection of finished components
- Documentation of material consumption, energy usage, and waste generation
- Assessment of process efficiency and recovery systems

The manufacturing process consists of two primary stages: stainless steel powder production via water atomization and component fabrication through Selective Laser Melting.

## 2.0 Process Description and Parameters

### 2.1 Powder Production via Water Atomization

The 316L stainless steel powder used in this process was produced through water atomization of X2CrNiMo1712 base material. This method involves melting the base alloy and disintegrating the molten stream with high-pressure water jets to form spherical powder particles.

**Key Process Parameters:** - Base material melting energy requirement: 2.23 MJ per kilogram of material - Atomization water usage: 16.8 kilograms - Electrical energy consumption for atomization process: 2.55 kWh

### 2.2 Selective Laser Melting Process

The SLM process utilized an EOS M290 system operating under argon atmosphere. The build chamber contained 33 flat washer components arranged in a single build plate configuration.

**Process Parameters:** - Total processing time: 13.38 hours - Laser-on power consumption: 5.5 kW - Laser-off/idle power consumption: 3.5 kW - Argon consumption per component: 54 liters - Initial chamber purge volume: 700 liters - Components per build cycle: 33 pieces

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## 3.0 Inspection Methods and Acceptance Criteria

### 3.1 Material Verification

Input materials were verified through certificate of analysis review and spot verification testing. The X2CrNiMo1712 stainless steel conforms to ASTM A240 specification for 316L composition.

### 3.2 Dimensional Inspection

Finished components underwent comprehensive dimensional verification using a Mitutoyo CMM (Coordinate Measuring Machine) with measurement uncertainty of  $\pm 0.005$  mm. Critical dimensions were measured according to the engineering drawing WSH-316L-20x10.5x2.0.

**Acceptance Criteria:** - Outer diameter:  $20.0 \text{ mm} \pm 0.1 \text{ mm}$  - Inner diameter:  $10.5 \text{ mm} \pm 0.1 \text{ mm}$   
- Thickness:  $2.0 \text{ mm} \pm 0.05 \text{ mm}$  - Surface roughness:  $R_a \leq 10 \text{ }\mu\text{m}$

### 3.3 Process Monitoring

Energy consumption, gas usage, and material flows were monitored through integrated sensors and manual logging. All instrumentation was calibrated within the past 6 months per calibration schedule CAL-AM-2023.

### 3.4 Visual Inspection

Components underwent 100% visual examination under 10x magnification for surface defects, including cracks, porosity, and dimensional abnormalities.

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## 4.0 Inspection Results

### 4.1 Material Consumption and Product Output

| Material Category     | Specific Item                | Quantity | Unit   | Verification Method             |
|-----------------------|------------------------------|----------|--------|---------------------------------|
| Raw Material Input    | X2CrNiMo1712 Stainless Steel | 4.11     | kg     | Material certificate & weighing |
| Process Media         | Water for Atomization        | 16.8     | kg     | Flow meter reading              |
| Process Gas           | Argon for SLM                | 3.08     | kg     | Calculated from volume usage    |
| Finished Product      | 316L Flat Washers            | 33       | pieces | Physical count                  |
| Finished Product Mass | 316L Flat Washers            | 0.61     | kg     | Precision scale measurement     |

### 4.2 Energy Consumption

| Process Stage     | Energy Type    | Consumption | Unit | Measurement Method    |
|-------------------|----------------|-------------|------|-----------------------|
| Powder Production | LV Electricity | 2.55        | kWh  | Utility meter reading |
| SLM Process       | LV Electricity | 64.92       | kWh  | Machine power monitor |

*Note: Historical data from previous quarter showed average SLM energy consumption of 68.2 kWh for similar component batches.*

### 4.3 Material Recovery and Reuse

| Recovery Stream | Material      | Quantity | Unit | Disposition                 |
|-----------------|---------------|----------|------|-----------------------------|
| Internal Reuse  | 316L Powder   | 2.94     | kg   | Recycled within SLM process |
| External Return | 316L Powder   | 0.15     | kg   | Returned to powder supplier |
| Water Recovery  | Process Water | 16.4     | kg   | Reclaimed for reuse         |

### 4.4 Waste Generation

| Waste Category | Material            | Quantity | Unit | Disposal Method |
|----------------|---------------------|----------|------|-----------------|
| Solid Waste    | Atomization Residue | 0.41     | kg   | Landfill        |
| Powder Waste   | Non-recyclable 316L | 0.01     | kg   | Landfill        |

### 4.5 Dimensional Verification Results

| Component ID | Outer Diameter (mm) | Inner Diameter (mm) | Thickness (mm) | Surface Ra (µm) | Status |
|--------------|---------------------|---------------------|----------------|-----------------|--------|
| WSH-001      | 20.01               | 10.51               | 2.01           | 8.2             | PASS   |
| WSH-002      | 19.99               | 10.49               | 1.99           | 7.8             | PASS   |
| WSH-003      | 20.02               | 10.52               | 2.02           | 8.5             | PASS   |
| ...          | ...                 | ...                 | ...            | ...             | ...    |
| WSH-033      | 20.00               | 10.50               | 2.00           | 8.1             | PASS   |

All 33 components met dimensional specifications. Complete measurement data available in Appendix A.

### 4.6 Material Property Verification

Chemical composition analysis confirmed all finished components meet 316L stainless steel specifications with chromium content of 16.5-18.5%, nickel 10-14%, and molybdenum 2-3%. Density measurements averaged 7.92 g/cm³, within the expected range for fully dense SLM components.

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## 5.0 Process Efficiency Assessment

The manufacturing process demonstrated effective material utilization with approximately 71.5% of input powder directly converted to finished components or reused within the process. The remaining material was either recovered for external recycling or accounted for as process waste.

**Material Balance Summary:** - Total stainless steel input: 4.11 kg - Finished product mass: 0.61 kg (14.8% yield) - Internal powder reuse: 2.94 kg (71.5% recovery) - External recycling: 0.15 kg (3.6% recovery) - Process losses: 0.41 kg (10.0% waste)

Industry benchmark for similar SLM processes typically shows material utilization between 65-75% for small component production.

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## 6.0 Non-Conformance and Corrective Actions

No non-conformances were identified during this inspection. All process parameters remained within specified ranges, and finished components met all quality requirements.

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## 7.0 Conclusions and Recommendations

### 7.1 Conclusions

The manufacturing batch of  $33 \times 316\text{L}$  stainless steel flat washers has successfully passed all quality inspection criteria. The Selective Laser Melting process operated within specified parameters, and all finished components conform to dimensional and material specifications.

### 7.2 Recommendations

1. Continue monitoring argon consumption trends to identify potential efficiency improvements in gas usage.
2. Investigate opportunities to reduce the 0.41 kg of solid waste from water atomization through process optimization.
3. Maintain current powder handling procedures to preserve the high material recovery rate achieved in this batch.

**Inspection Approval:**

Maria Rodriguez, Senior Quality Engineer

Date: October 26, 2023

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## Appendix A: Supplemental Data

Complete dimensional measurement records and material certification documents are maintained in the quality department records under reference QIR-AM-316L-231026-ATT.

**Document Revision History:**

Version 1.0 - Initial Release (October 26, 2023)